ASSIGNMENT

(in the course: Internet Application Technologies of the 6th semester of studies of the Department of Informatics & Telecommunications)

Title: Professional Networking Application (LinkedIn-style)

Introduction

The goal of this assignment is the development of a professional networking application. Users will have access to the application through a modern web browser.

In the application there are 2 roles: Administrator / Professional. The administrator role is assigned to a specific user / users during the installation of the application. Each role corresponds to a separate graphical user interface through which: A) the administrators manage the users of the application and export their data, B) the professionals manage the information about their education and their professional experience, make connection requests to other professionals and accept or reject connection requests, view and manage the network of professionals connected to them, post articles that can contain images and videos, post and respond to job ads, see on their timeline the articles posted by the professionals connected to them having the ability to comment on them, mark their interest in specific posts and are notified about the interest of other users in their posts, conduct private discussions with the professionals connected to them, navigate to the presentation pages of other professionals, and manage their account settings.

Application Requirements

1. The application initially displays a welcome page in the visitor's web browser. This page should give the user the ability to register in the application by pressing the corresponding button in the navigation bar. Also, it should give a user the ability to log in to the application by giving their email address and their user password in the navigation bar. The corresponding HTTP requests must be encrypted through the SSL/TLS protocol.
2. The registration page of a new user will require them to enter a user password, user password confirmation, first name, last name, email address, contact phone number, photo and any other details deemed necessary. If there is a user with the same email address, the application will display an appropriate message and the registration process will be terminated.
3. The application will have a pre-installed user who will have the role of the administrator. When the administrator correctly enters the email address that corresponds to them and their password they will be directed to the administration page.
4. Through the administration page the administrator should be able to navigate to the list of users. From the list of users they will be able to navigate to the details page of each user, through which they will be able to examine the details of the users. Also, the administrator will have the ability to export in XML format and JSON format the data of the users by selecting one or more of them (the résumé, the posts of articles and job ads, the professional experience, the expressions of interest (likes) and the comments they have posted on other professionals' articles, the network of the professionals connected to them).
5. After a user logs in to the application they will be navigated to the home page. At the top of the home page there will be a bar with the following options: Home Page / Network / Job Ads / Discussions / Notifications / Personal Details / Settings in the form of active or inactive tabs. The left part of the page contains a panel from where the user can navigate to their personal details as well as to their network. In the central part of the page the timeline of the professional is displayed, where the articles posted by the user and by the professionals connected to them are displayed in chronological order, on which they can add comments or mark their interest. Also displayed are articles of professionals not connected to them, for which users who belong to their network have marked their interest. Finally, a text and

multimedia (photos, video and audio files) input box is included, where they can post a new article.

6. By selecting the «Network» tab the user navigates to the Network page, where the professionals connected to them are displayed in a grid format (full name, photo, professional position, employer). By selecting with the mouse one of the connected professionals, the user navigates to the page of their personal details. From this page they can start a private discussion with that connected professional, in which case they navigate to the corresponding tab. Also there is a text input box for searching professionals who have registered in the application. The search results are displayed in a list, from where the user can navigate to the page of their personal details. In the case of a non-connected professional, the user cannot navigate to their network and sees only the information that the non-connected professional has marked as public.
7. By selecting the «Job Ads» tab the user sees the job ads that have been published by the professionals connected to them and can choose to apply for the job positions these ads refer to. Also they can post their own job ads and examine the applications that other professionals have made to them. In the «Job Ads» tab job ads that have been published by professionals not connected to them are also displayed. Job ads are displayed based both on the skills that the user has declared in their personal details **[Bonus]** and on a Matrix Factorization Collaborative Filtering algorithm that will process the data of the users' network. The vectors that are constructed represent the users in the space of the job ads. The values of the coordinates are determined by the views of the job ads by the users.
8. By selecting the «Discussions» tab the list of private discussions that the user has started with other professionals is displayed in the left part of the page. In the central part of the page the messages that have been exchanged in the most recent discussion or correspondingly in the one that the user will select from the list on the left are displayed.
9. By selecting the «Notifications» tab a page is displayed that contains two parts: the upper part concerns the connection requests which the user can accept or reject, as well as links through which they can navigate to the personal information page of those users. The lower part of the page displays the expressions of interest (likes) that other users have made on the user's article posts, along with the comments they have submitted.

10. By selecting the «Personal details» tab a page is displayed where the user can enter their professional experience, the information about their education and their skills. They can also mark which of this information is public and which private.
11. By selecting the «Settings» tab a page is displayed where the user can change their email address and their login password for the application.
12. **[Bonus]** In the user's timeline, articles are displayed based on a recommendation algorithm that will process the history of the expressions of interest (likes) of a user on articles of other users, the submission of comments on them as well as the network of the users. For the generation of the recommendations a Matrix Factorization algorithm will be used. Both in this question and in question 7 artificial datasets must be constructed since no data are freely available. In case a user has not marked interest and has not written any comment, the application will record and will use the views of the articles in order to construct the vectors that represent the users in the space of the articles. Articles whose authors belong to the network of a user will be displayed with priority. **The algorithms for generating recommendations must be implemented from scratch. The use of ready-made libraries is not allowed**.

Other Information

1. The assignment is mandatory and can be carried out either individually, or by groups of at most two (2) people. Its percentage of the total grade of the course is set at 80%. The Bonus questions can be implemented by the groups of **2 people** so that their members are exempted from participating in the written exam of the course.
2. The assignments will go through a plagiarism check. In case copying is found between two or more groups, all the involved assignments will automatically be graded with zero.
3. The assignment will be submitted electronically, exclusively through the e-class website of the course. The submission will include
    - a .zip file which will contain the source code and the necessary libraries,
    - a text file (in pdf), in which the design decisions, the assumptions, as well as various other details of implementation, installation and execution of the assignment are described. The structure of this file should be the following:
        - Cover page with the title of the assignment, the names of the group members and their student ID numbers
        - Table of contents,
        - Introductory chapter (which will briefly describe the goal of the assignment and the content of the chapters that follow),
        - Remaining chapters
        - Epilogue, where you will summarize the work during the development of the application, any difficulties you encountered and the way you dealt with them etc.
4. The assignment must be submitted according to the above way either by 14 July, 23.59 or from 15 July up to and including 20 September 2024, 23.59.
5. It should be emphasized that the design and the implementation of the application has a great degree of freedom, from the presentation (design of the web pages) and the design of the database to the addition of extra functionalities beyond those mentioned in the assignment

description. The quality of the organization of the code and of the design of the database will be graded according to what is taught in the course but also according to the general design principles of databases and world wide web applications.

6. **The Backend of the application must provide a REST API service programming interface. The Frontend of the application must be implemented using a JavaScript/TypeScript web framework (Angular, React, etc.). The services may be consumed only after the user's identity has been authenticated and based on authorization according to their role.**

Good Luck!